# Panagiotis Zazos

+30 698 371 7939 | pzazos@hotmail.com | linkedin.com/in/panagiotis-zazos-bala02188 | github.com/zazos

## Summary

Machine Learning Engineer with an applied-research background spanning explainable AI, large language models, and production deployment. EU-funded xAI research (MANOLO project; ExpliMed 2025 publication) grounds hands-on delivery across retrieval-augmented generation (RAG) pipelines, multi-agent LLM orchestration, model fine-tuning (PEFT/LoRA), and automated evaluation harnesses. Focused on turning research ideas into measurable, shippable systems—equally at home with literature and prototypes as with deployed, tested code.

## Technical Skills

**Programming**: Python, SQL, Ruby on Rails

**Machine Learning & Deep Learning**: PyTorch, TensorFlow, Keras, Scikit-learn, CNNs, GANs, Transfer Learning, Computer Vision

**LLM & Agentic Systems**: Large Language Models (LLMs), LangChain, LangGraph, Retrieval-Augmented Generation (RAG), ChromaDB, Anthropic Claude API, Multi-Agent Orchestration, Prompt Engineering, LLM-as-a-Judge

**Explainability & Evaluation**: Explainable AI (SHAP, LIME, Grad-CAM), Fine-tuning (PEFT / LoRA), DeepEval, Model Evaluation Metrics

**Libraries & Tools**: Pandas, NumPy, Hugging Face, Git, Docker, Tableau, D3.js, Prodigy

## Experience

**Data Scientist & Research Associate** — Oct 2024 – Present
*Four-Dot Infinity* — *Athens, GR*

- Task Leader for Explainability and Robustness within the EU-funded MANOLO research project, owning the workstream end to end.
- Designed and implemented a suite of explainable-AI techniques (SHAP, LIME, Grad-CAM) and authored novel evaluation metrics (fidelity, surrogacy efficacy score) to quantify and validate model interpretability.
- Translated complex technical solutions into European funding proposals, producing both technical specifications and non-technical narratives for mixed stakeholder audiences.
- Co-authored peer-reviewed academic papers based on project research, disseminating findings to the scientific community.

**Software Developer** — Jun 2021 – May 2022 & Jun 2023 – Mar 2024
*HQEngine* — *Melbourne, Australia (Remote)*

- Built features for a fintech start-up focused on Sales & Spend eInvoicing and ePayments using Ruby on Rails, HTML, and CSS.
- Integrated XML document generation to meet PEPPOL e-Invoicing standards for compliant digital transactions.
- Implemented validation mechanisms for Australian Business Numbers (ABNs) to ensure transaction legitimacy.
- Performed full-stack debugging to resolve issues and ensure reliable, regulation-compliant functionality.

**IT on ERP Entersoft Business Suite** — Jul 2022 – May 2023
*Hellenic Army* — *Athens, Vyronas, GR*

- During mandatory military service, used the Entersoft Business Suite ERP at the Special Army Unit Supply Center (EKEMS) to manage invoice recording, product codes, and commercial transactions.

**Informatics Intern** — Sep 2020 – Jan 2021
*Elxis Group* — *Athens, Illisia, GR*

- Built real-time tweet collection and filtering from the Twitter API based on earthquake events.
- Implemented tweet pre-processing and annotation with Prodigy to train a multi-class NLP model predicting earthquake intensity.

## PROJECTS

**Automated Pipeline for xAI-Driven Narrative Generation & Validation** LLM Project
*Local Development* *Python, LLMs, PEFT, LoRA, DeepEval, SHAP*

- Engineered an xAI module to extract feature attributions (SHAP, Grad-CAM) from target models into structured JSON payloads.
- Developed a 'Narrator' LLM, fine-tuned with PEFT/LoRA, to transform xAI outputs into domain-specific, compliant natural-language explanations.
- Implemented a 'Judge' LLM using DeepEval's GEval for automated validation of narrative quality against accuracy, safety, and actionability pillars.

**Conversational RAG System with Automated Evaluation Pipeline** Personal Project
*RAG, LangGraph, LLM Evaluation* *Python, LangChain, LangGraph, ChromaDB, Gemini*

- Built a RAG pipeline over a structured corpus using ChromaDB for vector retrieval and LangGraph for agent state management; deployed as a live Streamlit application.
- Implemented a dual-mode LLM backend supporting both cloud (Gemini API) and local (LM Studio) providers with zero code changes.
- Designed an LLM-as-a-Judge evaluation harness with four independent metrics—context precision (0.59), faithfulness (0.78), answer relevance (0.86), and correctness (0.75)—benchmarked against a hand-crafted 20-question ground-truth dataset.

**Multi-Agent System Design with Claude Code** Personal Project
*Agent Orchestration, Claude Code* *Python, Anthropic Claude Code, Markdown*

- Architected a three-agent pipeline using Claude Code primitives (slash commands, subagents, skills) with strict role separation across narration, rules-interpreter, and forked-context agents.
- Offloaded all arithmetic to an external Python rules engine, eliminating LLM computation errors and making business logic independently testable.
- Applied progressive skill loading to minimize context-window usage—each agent loads only the rule sections relevant to the current state.

**Music Mood Classification and Recommendation API** Master's Thesis
*Deep Learning, MIR, Web API* *Python, PyTorch, LoRA*

- Fine-tuned a pre-trained ResNet-18 on the MTG-Jamendo dataset using LoRA adapters for music mood classification.
- Applied transfer learning on extracted embeddings to an MLP for valence-arousal prediction against Spotify ground-truth data.
- Delivered a web API serving personalized recommendations with interactive visualizations of acoustic features and emotion mappings.

**Time-Series EEG Sleep Staging Classification** HORIZON European Project
*Deep Learning, Time-Series, XAI* *Python, TensorFlow, GANs*

- Built a CNN-CNN framework to classify 5 sleep stages from EEG epochs, using a custom focal loss to manage class imbalance.
- Engineered a style-transfer GAN in collaboration with NCSR Demokritos to transform EEG data, providing explainable insight into class-specific signal characteristics.

## PUBLICATIONS

**xSTAE: Explaining Classifier Decisions via Style Transfer Autoencoding** Bologna, Italy
*ExpliMed 2025* *2025*

Co-authored paper introducing a style-transfer system for counterfactual explanations in sleep-stage classification.

**Contribution:** Designed and implemented the dual-CNN classification framework used as the backbone for the explainer system.

## EDUCATION

**National and Kapodistrian University of Athens** Athens, GR
*M.S. in Data and Knowledge Management* *Sep 2023 – Feb 2026*

**National and Kapodistrian University of Athens** Athens, GR
*B.S. in Informatics and Telecommunications* *Jan 2015 – Sep 2021*

## LANGUAGES & INTERESTS

**Languages**: Greek (Native), English (Highly Proficient), Japanese (N5)

**Interests**: CrossFit training, Reading, Writing, Music, Video Games